

Python: Language & Basic Syntaxes

Introduction to Python

Python is a high-level, interpreted, and general-purpose programming language created by Guido van Rossum and released in 1991. Known for its clean and readable syntax, Python is widely used in web development, data science, artificial intelligence, automation, and scripting. Its philosophy emphasises code readability and simplicity, making it an ideal first language for beginners.

Variables and Data Types

Python variables do not require explicit type declarations. The interpreter infers types automatically.

Example:

```
name = 'Alice' # String
age = 25 # Integer
height = 5.6 # Float
is_active = True # Boolean
```

Python supports several built-in data types including str, int, float, bool, list, tuple, dict, and set.

Control Flow

Python uses indentation to define code blocks instead of curly braces.

If-Else:

```
if age >= 18:
    print('Adult')
else:
    print('Minor')
```

For Loop:

```
for i in range(5):
    print(i)
```

While Loop:

```
while age > 0:
    age -= 1
```

Functions

Functions are defined using the def keyword.

```
def greet(name):
    return 'Hello, ' + name
```

```
print(greet('Alice')) # Output: Hello, Alice
```

Python also supports default arguments, `*args` (variable positional arguments), and `**kwargs` (variable keyword arguments), offering great flexibility in function design.

Lists, Tuples, and Dictionaries

List (mutable): `fruits = ['apple', 'banana', 'cherry']`

Tuple (immutable): `coords = (10.5, 20.3)`

Dictionary (key-value pairs): `person = {'name': 'Alice', 'age': 25}`

Lists support methods like `append()`, `remove()`, and `sort()`. Dictionaries allow fast lookups by key and are widely used for structured data.

Modules and Libraries

Python's vast ecosystem of libraries is one of its greatest strengths. Standard library modules like `os`, `sys`, `math`, and `datetime` are built-in. Third-party libraries such as NumPy (numerical computing), Pandas (data analysis), Flask (web development), and TensorFlow (machine learning) can be installed using `pip` — Python's package installer. Importing a module is as simple as writing: `import math` or `from datetime import date`.